

Impact of Dietary Patterns on Postprandial Hyperglycemia

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Abstract

Dietary habits are one of the major factors for the rapidly rising incidence of type 2 diabetes and higher glucose variability with marked daytime postprandial hyperglycemia is significantly associated with frailty in older patients with diabetes. Therefore, in this final project, we aim to investigate the effect of dietary patterns on postprandial hyperglycemia in diabetes patients based on a publicly available dataset, CGMacros. The outcome of interest is the time to the first occurrence of postprandial hyperglycemia, which is defined as an estimated blood glucose greater than 180 mg/dL (10 mmol/L) according to CGM readings. Covariates include meal type (lunch or dinner), macronutrients, glucose at the beginning of each meal, and meal time. We used survival analysis techniques to explore relationships between dietary factors and postprandial hyperglycemia events. Our findings suggest that the effect of gender on postprandial hyperglycemia risk is opposite for individuals with and without diabetes, with males having a lower risk in the diabetes group but a higher risk in the non-diabetes group. Additionally, carbohydrate intake increases the risk of postprandial hyperglycemia over time.

1 Introduction

Type 2 diabetes (T2D) has become a major global health concern, with its prevalence increasing at an alarming rate. An estimated 462 million individuals worldwide are affected by T2D, representing approximately 6.28% of the global population (Khan et al., 2020). Postprandial hyperglycemia (PPH), defined as a rapid increase in blood glucose levels following a meal, appears to play a crucial role in the pathophysiology of late diabetes complications (Vlachos et al., 2020). Research indicates that glucose variability, especially daytime postprandial hyperglycemia, rather than average glucose levels alone, is a significant predictor of frailty in older adults (Chung et al., 2021). Given its role in disease progression, managing PPH effectively is essential for reducing the long-term health risks associated with diabetes.

Dietary intake plays a crucial role in the onset of PPH, and the progression of diabetes. Previous studies have shown that a diet rich in carbohydrates can contribute to insulin resistance and postprandial hyperglycemia, ultimately increasing the risk of developing diabetes (Sami et al., 2017). While carbohydrate intake is a well-established determinant of PPH, the impact of other macronutrients, such as proteins and fats, remains less clear.

Most previous research has been conducted in controlled laboratory settings, which may not fully capture the complexities of dietary habits in free-living conditions. To address this gap, this study utilizes a dataset which provides continuous glucose monitoring (CGM) and detailed dietary intake data from individuals over multiple days. By applying Cox proportional hazards models, we aim to investigate the time to first occurrence of postprandial hyperglycemia and examine the demographic information, lab test results, and dietary factors associated with increased risk.

2 Methods

2.1 Data Source

CGMacros is a dataset containing multi-modal information from CGM readings and food macronutrients. It has data from 45 study participants who consumed meals with varying and known macro-nutrient compositions in a free-living setting for 10 consecutive days (Gutierrez-Osuna et al., 2025). For those who have dietary records, we excluded their records during the first and the last day of follow-up to ensure the quality of CGM readings. We consider each meal record as an observation. We defined the event, postprandial hyperglycemia as the Dexcom G6 Pro CGM readings being greater than 180 mg/dL (10 mmol/L) after lunch or dinner. We are interested in the time to first occurrence of PPH for each meal. If a participant did not experience hyperglycemia within 3 hours postprandially or consumed food within these 3 hours, the data is considered as right censored. If a participant had a CGM reading greater than 180 mg/dL at the beginning of a meal (baseline glucose > 180 mg/dL), then this record was excluded from the analysis. The data filtering process is shown in Figure 1.

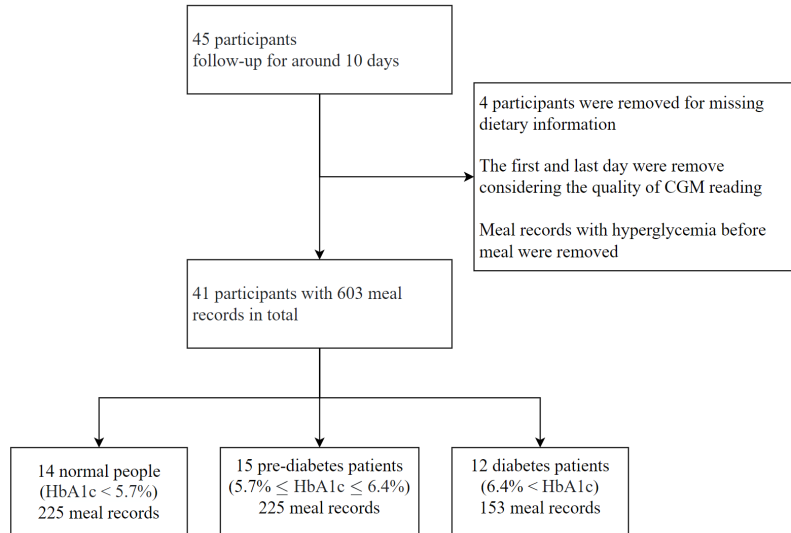


Figure 1: Participant and Meal Record Selection Flowchart.

The information on demographics and lab test results was collected at the beginning of the study. Then the participants wore CGM devices to collect their glucose every 5 minutes. Before each meal, the participants took a photo of their food and uploaded it, the information on macronutrients was automatically calculated. We further constructed a factor, late meal time, defined as the time difference from the average meal time (12:30 p.m. for lunch, and 6:30 p.m. for dinner), to capture the information on meal time.

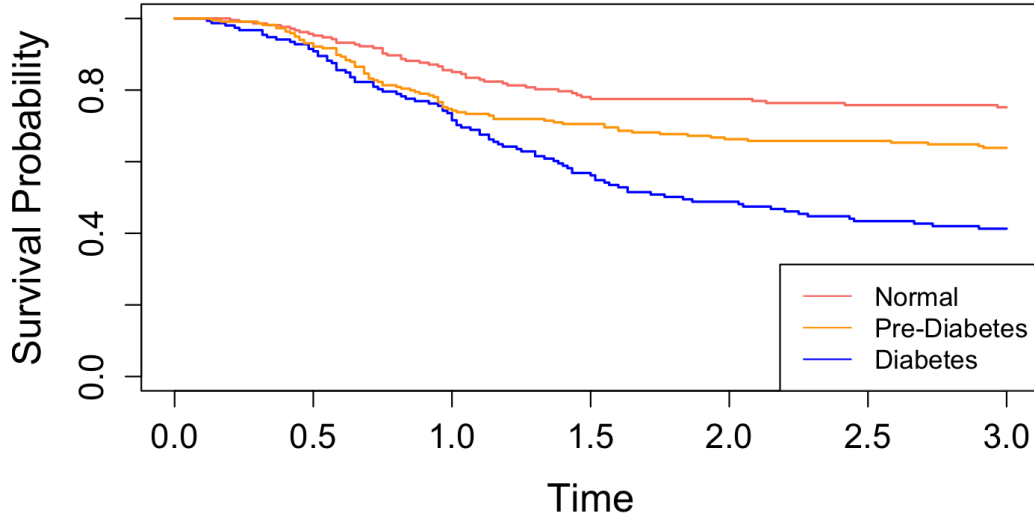


Figure 2: Kaplan-Meier survival curves comparing *Diabetes*, *Pre-Diabetes* and *Normal* groups without adjusting for covariates.

Figure 2 presents the KM curve for three groups. The *Normal* group has the highest survival probability, followed by the *Pre-Diabetes* group, and the *Diabetes* group has the least. Further exploration can be found in the next section.

To better model the hazard function, we incorporated covariates including demographic information, lab tests at the beginning of the study, and dietary factors. The first two are consistent within each subject, and the last one is different among different meal records.

2.2 Statistical Methods

Kaplan-Meier survival curves and Nelson-Aalen cumulative hazard curves were both generated to visualize how quickly participants experienced postprandial hyperglycemia after meals.

To explore the factors that increase the hazards of PPH. We constructed two Cox proportional hazards models, the first one took advantage of demographic information and baseline lab test results. The second model further explored the impact of dietary habits and macronutrients on PPH using stratified Cox regression.

For the first model, two curves and the subsequent first analysis are based on grouping by diabetes status. The strata do not satisfy the proportional hazards assumption, which was validated by Figure S1–Figure S2. We can see from the two figures that the strata *Normal* and *Pre-Diabetes* follow the PH assumption so we combine these two categories as a single group for further analysis. We categorized subjects with $A1C \leq 6.3$ as *Non-Diabetes* and $A1C > 6.3$ as *Diabetes*.

For the survival analysis, two separate Cox proportional hazards models were constructed. We included several baseline covariates, including (1) demographic characteristics: *Age*, *Gender*, *BMI*, *LDL*, *HDL*, and *Triglycerides*, and (2) meal data: *Baseline Glucose (centered)*, *Meal Type*, and *Calories*, and *Late Meal Time*. The Breslow method was applied to handle cases where multiple subjects had events at the same time point (ties). In the first analysis, all meal records were treated as independent observations without considering the repeated measurements from the same subject.

Finally, stepwise model selection based on the Akaike Information Criterion (AIC) was applied to both analyses. To further assess the goodness of fit, a Cox-Snell residual plot was generated.

The second analysis explicitly accounted for multiple meal observations per subject by including a stratification variable based on subject ID, thus controlling for within-subject correlation. Due to stratification, only meal-level covariates were incorporated in the second model. The covariates include *Baseline Glucose (centered)*, *Carbohydrate Intake*, *Protein Intake*, *Fat Intake*, *Fiber Intake*, *Meal Type* and *Late Meal Time*. Proportional hazard tests were conducted to test the assumption. If the test identified variables that violated the proportional hazards assumption, these variables were adjusted by incorporating time interactions in the model.

3 Results

3.1 Descriptive Analysis

Table 1: Baseline Characteristics of Study Participants by Diabetes Status

Term	Diabetes (n=12)	Normal (n=14)	Pre-diabetes (n=15)	p-value	
Age (years)	53.25 (5.50)	39.93 (14.29)	51.40 (11.01)	0.007	**
Gender (Male)	5	5	4	0.708	
BMI	31.64 (4.73)	28.65 (6.34)	32.99 (7.50)	0.191	
Triglycerides (mg/dL)	253.58 (295.91)	105.64 (64.31)	138.73 (72.03)	0.082	.
LDL (mg/dL)	116.91 (52.21)	108.57 (40.02)	118.40 (21.71)	0.771	
HDL (mg/dL)	44.50 (13.96)	60.00 (13.13)	48.93 (17.39)	0.033	*
HbA1c (%)	7.29 (0.60)	5.26 (0.31)	5.99 (0.22)	<0.001	***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, . $p < 0.1$

Data are presented in mean (SD) for continuous variables and n for categorical variables.

The p-values were given by the Kruskal-Wallis test (continuous variables) and the Chi-squared test (categorical variables).

Table 1 shows the demographic information and baseline lab test results of participants in three diabetes statuses. There are significant differences in age and high-density lipoprotein cholesterol among three groups. This indicates those covariates may have different effects on PPH in people at different diabetes stages. Table 2 presents the characteristics of 603 meal records.

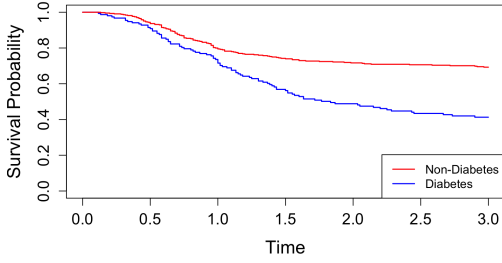
Table 2: Meal Characteristics of 603 Meal Records

Term	Mean (SD) or n (%)
Calories (kcal)	622.73 $\pm$ 339.54
Carbohydrate (g)	63.96 $\pm$ 45.12
Protein (g)	36.86 $\pm$ 25.38
Fat (g)	24.42 $\pm$ 25.99
Fiber (g)	34.55 $\pm$ 227.02
Baseline Glucose (mg/dL)	116.58 $\pm$ 22.61
Late Meal Time (hrs)	0.1 $\pm$ 1.45
Meal Type (Lunch)	328 (54.39%)

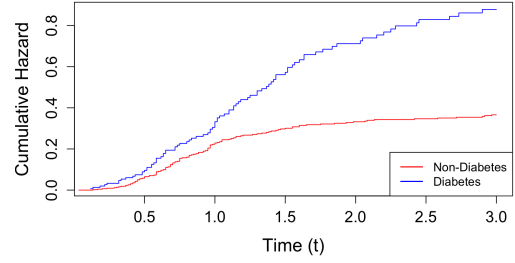
To further examine survival characteristics, Kaplan-Meier survival curves and Nelson-Aalen cumulative hazard curves were generated in Figure 3a–Figure 3b. We can see from both curves that the survival risk for diabetes subjects is much higher than that for the non-diabetes.

3.2 Cox Proportional Hazards Model Without Subject Stratification

We first fitted a Cox proportional hazards (PH) regression model that included baseline covariates without considering subject stratification or time-dependent effects and then



(a) Kaplan-Meier survival curves comparing the *Diabetes* and *Non-Diabetes* groups without adjusting for covariates.



(b) Nelson-Aalen cumulative hazard estimates comparing the *Diabetes* and *Non-Diabetes* groups without adjusting for covariates.

Figure 3: Log cumulative hazard plot for different diabetes status groups.

we used AIC to select variables. For each category of diabetes status, we fitted a separate model. Two models after AIC selection are shown in Table 3, including coefficients, relative risks, standard errors, and corresponding significance levels. The variables excluded by AIC selection are marked with “—” in the table.

For the *Diabetes* group, significant variables include *Gender*, *Triglycerides*, *LDL*, *HDL*, and *Meal Type*, with all these corresponding p-values < 0.001 . The hazard ratio for *Gender* is 0.2636, which means males have a significantly lower risk compared to females. The increase in *LDL* values is associated with an increased risk from the coefficient, while higher *HDL* levels are linked to a decreased risk.

For the *Non-Diabetes* group, there is a different set of significant variables, including *Age*, *Gender*, *LDL*, *HDL*, *Baseline Glucose (centered)*, *Meal Type*, and *Calories*. *Triglycerides* is excluded in the final model. The effect of *Gender* is opposite to that in the *Diabetes* group, with a hazard ratio of 1.5536 (p-value = 0.033), which means a significantly higher risk for males compared to females. *Baseline Glucose (centered)* becomes significant (p-value < 0.0001) with a hazard ratio of 1.0213, suggesting a significant association with postprandial hyperglycemia.

Table 3: Cox proportional hazards model summary for the Diabetes and Non-Diabetes groups after AIC-based variable selection.

Diabetes Group					
Term	Coeff	HR	SE	p-value	
Age (years)	0.0374	1.0382	0.0231	0.104	
Gender (Ref: Female)	-1.3335	0.2636	0.3080	<0.001	***
BMI	—	—	—	—	
Triglycerides (mg/dL)	-0.0112	0.9889	0.0021	<0.001	***
LDL (mg/dL)	0.0141	1.0142	0.0028	<0.001	***
HDL (mg/dL)	-0.0840	0.9194	0.0179	<0.001	***
Centered Baseline Glucose (mg/dL)	0.0074	1.0074	0.0050	0.143	
Meal (Ref: Dinner)	-1.0819	0.3390	0.2505	<0.001	***
Calories (kcal)	0.0008	1.0008	0.0004	0.061	
Late Meal Time (hrs)	—	—	—	—	
Non-Diabetes Group					
Term	Coef	HR	SE	p-value	
Age (years)	0.0215	1.0218	0.0078	0.006	**
Gender (Ref: Female)	0.4406	1.5536	0.2068	0.033	*
BMI	—	—	—	—	
Triglycerides (mg/dL)	—	—	—	—	
LDL (mg/dL)	-0.0060	0.9940	0.0028	0.031	*
HDL (mg/dL)	-0.0161	0.9840	0.0070	0.022	*
Centered Baseline Glucose (mg/dL)	0.0210	1.0213	0.0043	<0.001	***
Meal (Ref: Dinner)	-0.3215	0.7250	0.1840	0.081	.
Calories (kcal)	0.0007	1.0007	0.0002	0.001	**
Late Meal Time (hrs)	—	—	—	—	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, . $p < 0.1$

We also generated a Cox-Snell residual plot in Figure 4 to validate the appropriateness of these two final models, where the Nelson-Aalen cumulative hazard function of the Cox-Snell residuals closely followed the 45-degree line for the *Diabetes* group at all residual values while only at lower residual values but deviated at larger residual values for the *Non-Diabetes* group, suggesting two models fit reasonably well for most observations but may still need some improvement or covariates adjustments.

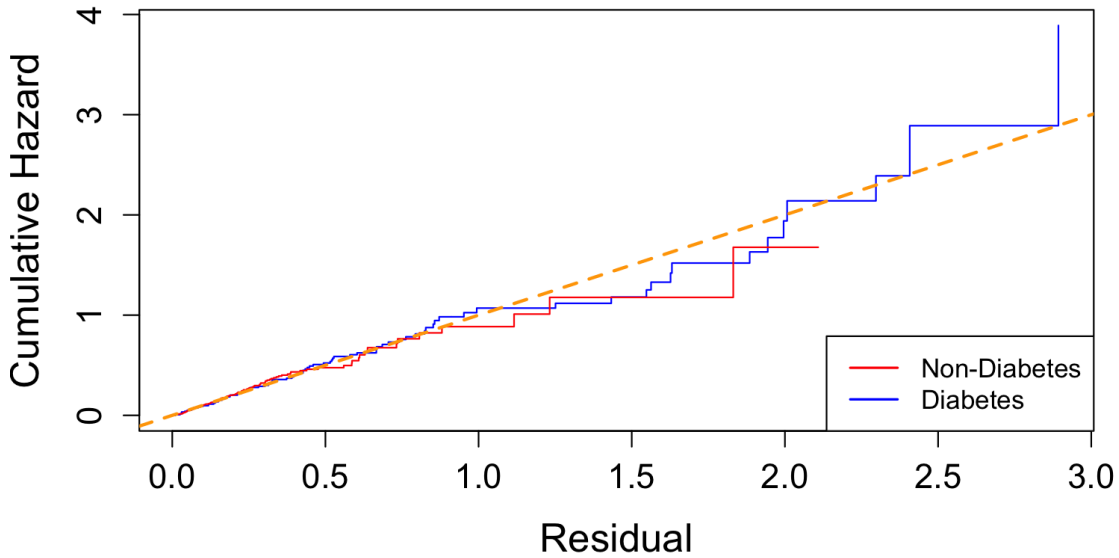


Figure 4: Cox-Snell residual plot for assessing model fit. The blue and red curves represent the Nelson-Aalen cumulative hazard function of the residuals for *Diabetes* and *Non-Diabetes* groups, respectively, while the orange dashed line corresponds to the expected cumulative hazard under a perfect model fit. A close alignment between the two indicates a well-fitted model.

3.3 Cox Proportional Hazards Model Stratified by Subject

To further explore the impact of dietary patterns and macronutrients for each meal on postprandial hyperglycemia, the Cox proportional hazards model stratified by subject was constructed to adjust within-subject variability. The result is given in Table S1.

The proportional hazards assumption for each covariate was tested using Proportional hazards tests (Grambsch and Therneau, 1994). According to the results in Table S2, baseline glucose and carbohydrate did not satisfy the proportional hazards assumption. Therefore, time interaction with those two covariates was added to address the violation of the assumption. According to the Schoenfeld residuals plots in Figure S3, the effect of baseline glucose decreases linearly with time, and the effect of carbohydrate intake increases linearly with time. Therefore, we add the interaction between centered baseline glucose and time, as well as the interaction between carbohydrate intake and time. The result is given below in Table 4. The Harrell’s C-index of the model is 0.725, indicating good discrimination ability. Prediction accuracy was measured by R^2 and L^2 (Li and Wang, 2016). The corresponding R^2 is 0.6908, and L^2 is 0.0491. This indicates that around 69% of the variability in the response is explained by a corrected prediction based on the original prediction function, but only 5% of the prediction error in the original prediction function is explained by the corrected prediction function.

Table 4: Results of Cox Proportional Hazards Model with Time Interaction Stratified by Subject

Term	Coef	HR	SE	p-value	
Centered Baseline Glucose (mg/dL)	0.0291	1.0295	0.0082	<0.001	***
Carbohydrate (g)	0.0034	1.0034	0.0037	0.357	
Protein (g)	-0.0067	0.9933	0.0036	0.060	.
Fat (g)	-0.0214	0.9788	0.0066	0.001	**
Fiber (g)	-0.0002	0.9998	0.0006	0.759	
Meal (Ref: Dinner)	-0.6240	0.5358	0.1674	<0.001	***
Late Meal Time (hrs)	0.1408	1.1512	0.0665	0.034	*
Centered Baseline Glucose $\times$ Time	-0.0180	0.9821	0.0066	0.006	**
Carbohydrate $\times$ Time	0.0142	1.1043	0.0038	<0.001	***

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, . $p < 0.1$

According to Table 4, the coefficient for baseline glucose is positive and significant and the coefficient of its interaction with time is negative. This is coincident with intuition since people with higher glucose levels before meals tend to reach hyperglycemia quickly, but this tendency will decrease as time increases. Initial carbohydrate intake may not significantly increase the hazards of PPH, while its effect becomes more pronounced as time progresses. This may indicate a delayed or cumulative effect of dietary carbohydrates on blood glucose regulation. In addition, protein and fat reduce the hazards of PPH to some extent. Another exploration is that the hazard ratio of lunch and dinner is 0.54, and late meal time will increase the hazards of being PPH, which gives insight that diabetes patients should eat less during dinner, and pay attention to punctual meals.

4 Discussion

The study found that carbohydrate intake alone does not cause an immediate large increase in blood sugar levels, but its effect becomes stronger over time. This finding may be important for people who have impaired glucose metabolism, such as those with prediabetes or type 2 diabetes, where delayed hyperglycemia can contribute to poor glycemic control and increased risk of diabetes complications. Therefore, careful management of carbohydrate consumption, particularly over extended periods, is critical for maintaining better glucose control in clinical practice.

Additionally, our findings suggest that other dietary factors, like protein and fat intake, may have prohibitive effects against rapid increases in blood glucose. Incorporating balanced meals with adequate protein and fat could potentially moderate glucose responses and help mitigate the negative impact of carbohydrates. Future dietary recommendations for diabetes management should be considered in several ways, such as emphasizing balanced nutrient intake alongside carbohydrate monitoring.

However, our study also has some limitations. One limitation of our study is the handling of subject-specific variability. Although we stratified by subject in the Cox proportional hazards model, we did not explicitly model subject-level random effects, which could more accurately account for individual variability through mixed-effects models. In the future, we will consider incorporating frailty models or other random effects methods to improve our studies.

5 Conclusion

In this report, we studied the factors that influence PPH using the CGM data and dietary intake records in a free-living environment. By applying Kaplan-Meier survival analysis and Cox proportional hazards models, we examined the impact of demographic characteristics, baseline metabolic markers, and macronutrient intake at the time of first PPH occurrence after a meal.

We applied a Cox proportional hazards model with and without subject stratification. For the model without subject stratification, we identified different risk factors for subjects with and without diabetes and found that the effect of gender on PPH risk was the opposite in the diabetes and non-diabetes groups. For within-subject variability, we employed a stratified Cox model and found that baseline glucose and carbohydrate intake influenced PPH risk dynamically over time. Protein and fat intake were also associated with reduced risk, and late meals increased the likelihood of PPH. These findings highlight the importance of balanced macronutrient intake and timely eating habits in managing blood glucose levels.

Supplementary Materials

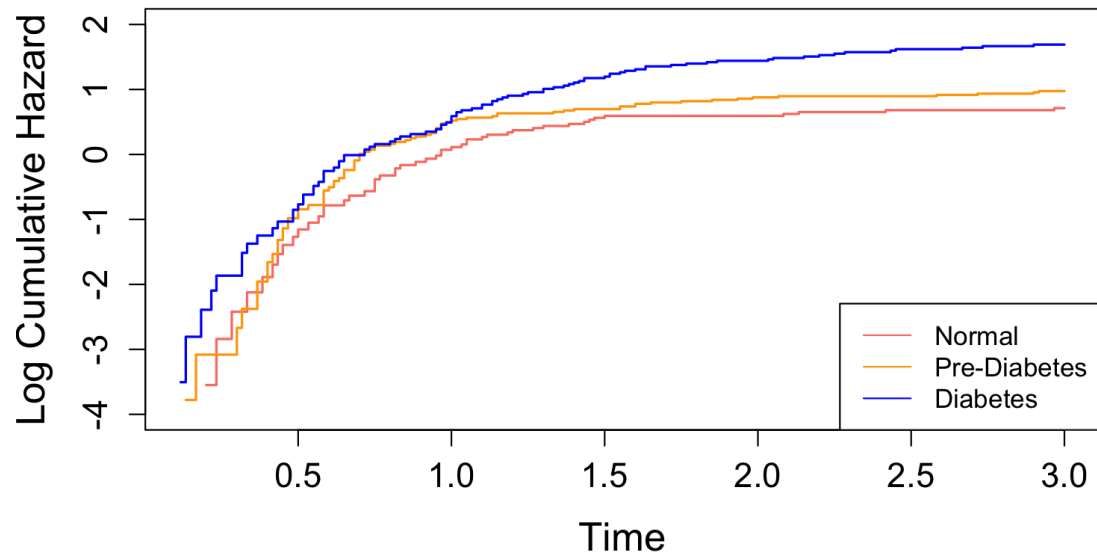


Figure S1: Log cumulative hazard estimates for different diabetes status groups.

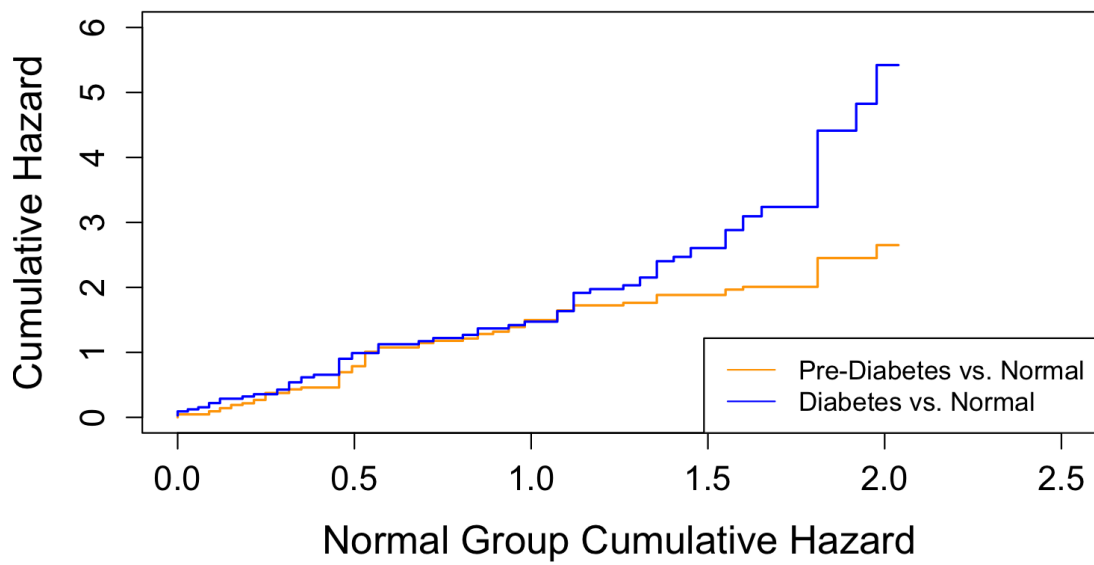


Figure S2: Andersen plot comparing the cumulative hazard functions for different diabetes status groups relative to the *Normal* group.

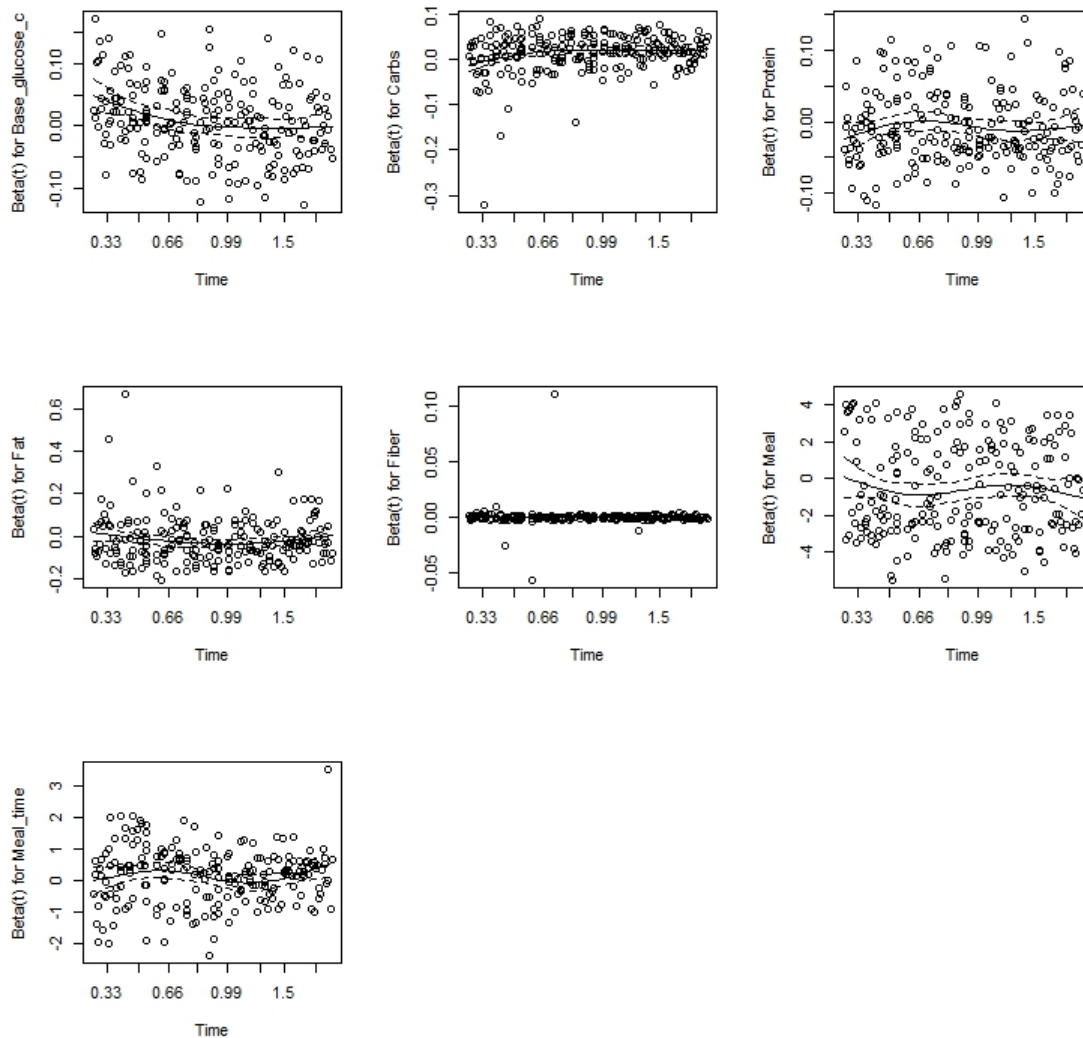


Figure S3: Schoenfeld Residuals Plots. Each panel represents a different covariate:

(1) Centered Baseline Glucose (top-left), (2) Carbohydrate Intake (top-center), (3) Protein Intake (top-right), (4) Fat Intake (middle-left), (5) Fiber Intake (middle-center), (6) Meal Type (middle-right), (7) Meal Time (bottom-left). The y-axis shows the scaled Schoenfeld residuals and the x-axis represents time. Deviations from a horizontal trend suggest a violation of the proportional hazards assumption.

Table S1: Results of Cox Proportional Hazards Model Stratified by Subject

Term	Coef	HR	SE	p-value	
Centered Baseline Glucose (mg/dL)	0.0090	1.0090	0.0041	0.029	*
Carbohydrate (g)	0.0143	1.0144	0.0023	<0.001	***
Protein (g)	-0.0079	0.9921	0.0035	0.025	*
Fat (g)	-0.0178	0.9823	0.0064	0.005	**
Fiber (g)	-0.0001	0.9999	0.0006	0.849	
Meal (Ref: Dinner)	-0.5731	0.5638	0.1647	<0.001	***
Late Meal Time (hrs)	0.1358	1.1454	0.0656	0.039	*

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, . $p < 0.1$

Table S2: Results of Proportional Hazards Tests

Term	χ^2	df	p
Centered Baseline Glucose (mg/dL)	13.06	1	<0.001
Carbohydrate (g)	16.56	1	<0.001
Protein (g)	0.18	1	0.674
Fat (g)	3.21	1	0.073
Fiber (g)	0.03	1	0.869
Meal (Ref: Dinner)	0.35	1	0.552
Late Meal Time (hrs)	0.00	1	0.996
GLOBAL	29.88	7	<0.001

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